IIDS67692 Computational Methods for Multimodal Data Analysis

Student ID: 14210037

Part A: (word counts: 498)

Methods

Spatial Graph Convolutional Network (SpaGCN) is a graph-based deep learning framework for spatial transcriptomics analysis (Hu et al., 2021). It performs **early fusion** by combining three modalities — gene expression counts, spatial coordinates, and histology image pixel intensities — prior to any learning step. These signals are merged during graph construction: a weighted adjacency matrix encodes both Euclidean distance between tissue spots and morphological similarity derived from histology, with parameter *s* controlling the relative image contribution. Because all modalities are integrated before feature learning begins, the architecture constitutes an early fusion design, enabling each modality to directly shape the GCN latent representation. The fused graph is then processed by GCN layers and refined through iterative deep clustering (DEC) initialised by k-means.

The Visium DLPFC dataset (sample 151676) was loaded from filtered_feature_bc_matrix.h5, spatial barcodes, and the full-resolution histology image (151676_full_image.tif). Preprocessing involved extracting RGB pixel intensities at each spot's coordinate and constructing the spatial weight matrix. SpaGCN was applied under *s* = 0, 1, and 10, followed by k-means clustering with 7 clusters matching the ground-truth labels (Layers 1–6 and White Matter). Clusters were visualised spatially and evaluated using ARI, FMI, AMI, and per-label F1-scores.

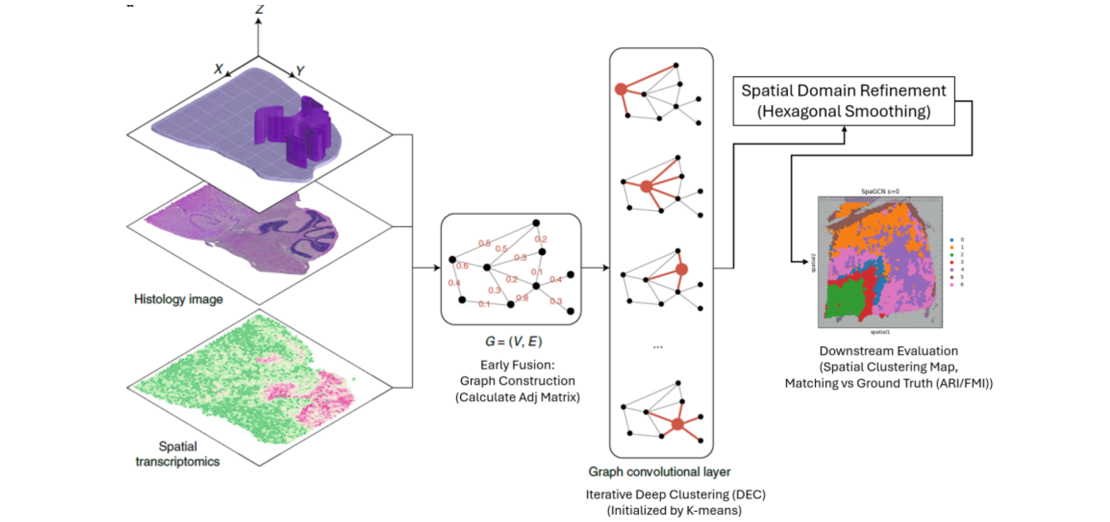


Figure 1. Overview of the SpaGCN Workflow.

Gene expression, histology, and spatial coordinate data undergo early fusion to build an adjacency graph. The integrated features are processed via graph convolutional layers and refined through hexagonal smoothing to delineate spatial domains. The final spatial clustering maps are utilised for downstream evaluation by benchmarking and matching against the annotated ground truth.

Results

Across all *s* values, Layer 1, White Matter (WM), and Layer 6 are consistently identified, as these structurally extreme regions are robustly anchored by both transcriptomic and morphological signals. Critical differences emerge in intermediate cortical layers (2–5).

At **s = 0**, transcriptomics-only analysis produces the highest global alignment (FMI = 0.3648, AMI = 0.3838), sharply separating WM (F1 = 0.84) and Layer 1 (F1 = 0.68). However, without morphological guidance, intermediate boundaries become noisy; Layer 4 is entirely missed (F1 = 0.00) because its transcriptomic signature is insufficiently distinct from adjacent layers. Convergence is rapid at 46 epochs.

At **s = 1**, moderate histological input acts as a spatial smoothing constraint, suppressing sequencing noise while preserving biological continuity. The model achieves the most anatomically faithful stratification of intermediate layers, though global metrics are moderate (ARI = 0.2105, FMI = 0.3457). WM (F1 = 0.76) and Layer 1 (F1 = 0.65) remain well recovered, but Layer 4 is again undetected (F1 = 0.00).

At **s = 10**, image features overwhelm transcriptomic signals, producing over-smoothed cluster boundaries and the lowest global metrics (FMI = 0.3425, AMI = 0.3564). Layer 1 (F1 = 0.43) and Layer 6 (F1 = 0.22) degrade as morphological entanglement absorbs adjacent spots. Critically, *s* = 10 is the only configuration recovering Layer 4 (F1 = 0.21), demonstrating that heavy histological weighting provides the texture-level contrast necessary to resolve this elusive stratum. GCN optimisation is hampered, requiring 121 epochs to converge.

Conclusion

No single *s* value is universally optimal. *s* = 0 maximises global accuracy for macro-structures; *s* = 1 provides the most anatomically faithful spatial continuity; *s* = 10 uniquely resolves subtle mid-layer boundaries at the cost of broader coherence. Parameter selection should be guided by clinical objectives: transcriptomics-only for structurally distinct boundaries, heavy histological weighting for granular mid-layer resolution.

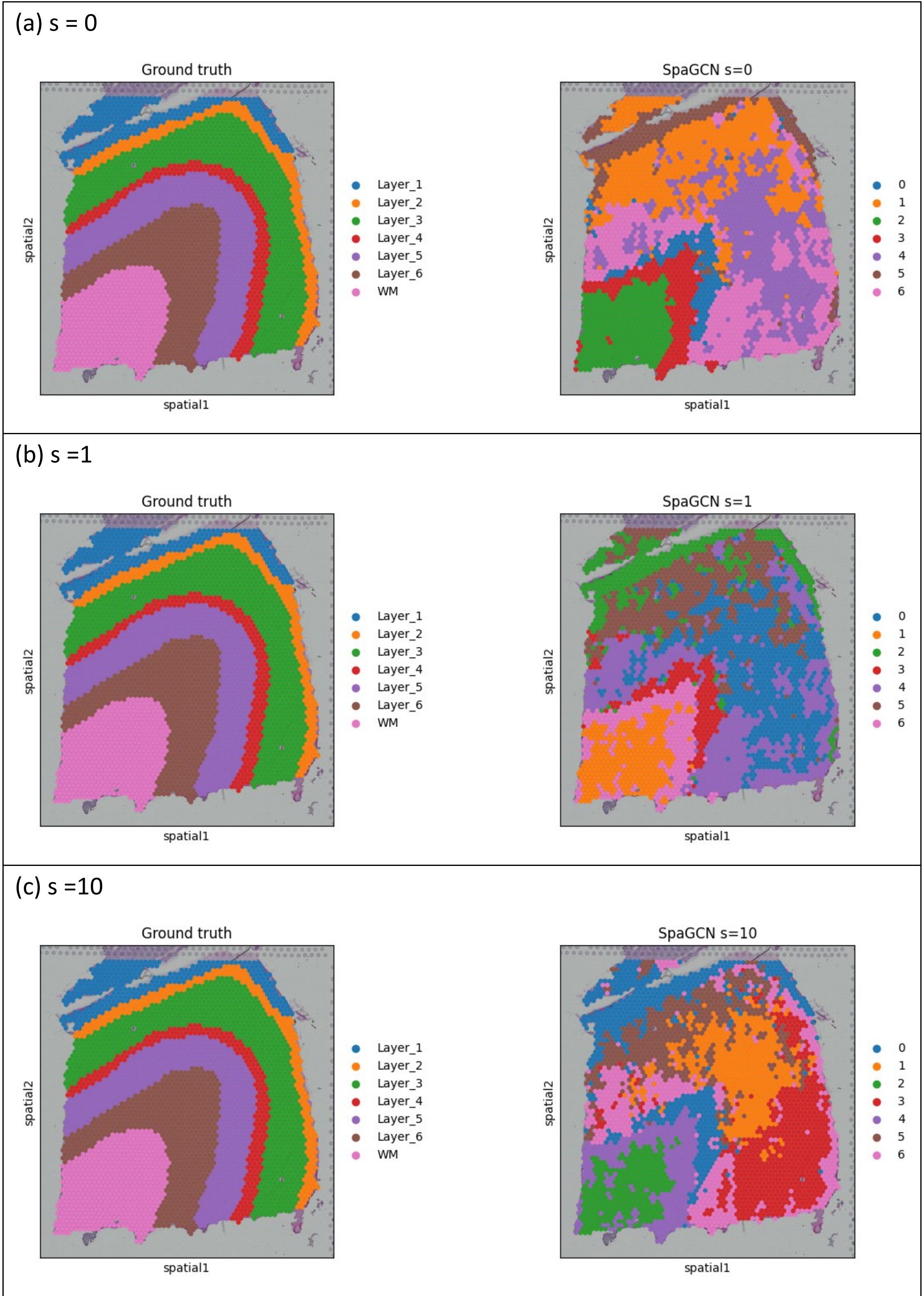


Figure 2. Spatial Clustering Maps Across Different Histology Contribution Settings. The visual comparison illustrates how the histology weight parameter s=0, s=1, and s=10 affects spatial domain identification and cortical layer stratification, benchmarking the predicted spatial clusters against the manually annotated ground-truth map.

Student ID: 14210037

Table 1. Quantitative Evaluation and Per-Label Classification Performance Under Varying Histology Weights s.

The table presents global clustering metrics (FMI, ARI, AMI) and granular layer-by-layer classification reports (Precision, Recall, F1-score) to evaluate the impact of the image contribution parameter s = 0, 1, 10 on SpaGCN.

	s = 0			s = 1			s = 10			
	FMI	ARI	AMI	FMI	ARI	AMI	FMI	ARI	AMI	
	0.3648	0.2293	0.3838	0.3457	0.2105	0.3677	0.3425	0.2171	0.3564	
Layer	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Support
Layer_1	0.71	0.66	0.68	0.56	0.77	0.65	0.43	0.82	0.56	289
Layer_2	0.1	0.3	0.15	0.1	0.3	0.15	0.2	0.44	0.27	254
Layer_3	0.5	0.49	0.49	0.52	0.39	0.45	0.55	0.35	0.43	836
Layer_4	0	0	0	0	0	0	0.15	0.37	0.21	254
Layer_5	0.39	0.47	0.43	0.38	0.47	0.42	0.51	0.4	0.45	649
Layer_6	0.92	0.26	0.41	0.91	0.28	0.43	0.3	0.17	0.22	616
WM	0.99	0.72	0.84	0.98	0.62	0.76	1	0.56	0.71	533

Student ID: 14210037

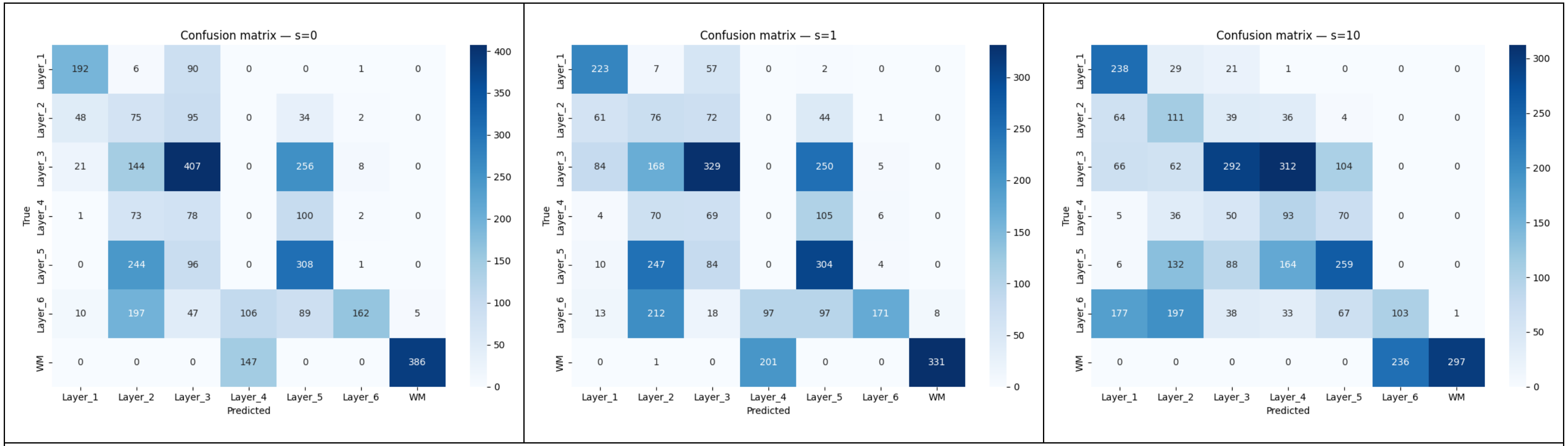


Figure 3. Confusion Matrices for SpaGCN Clustering Across Varying Image Weights.

The plots display the spot-level classification accuracy for s = 0, s = 1, and s = 10. Mechanistically, the purely transcriptomic approach s = 0 demonstrates strong performance in isolating Layer 3. Conversely, heavily weighting the histology s = 10 significantly improves the resolution of intermediate cortical layers; most notably, it serves as the exclusive configuration capable of detecting Layer 4, which remains entirely unresolved (F1-score of 0.00) under the other two settings.

Student ID: 14210037

Part B (word counts: 483)

Dataset and Preprocessing

The PathVQA dataset for Medical Visual Question Answering (VQA) consists of pathology images and question-answer pairs. After removing unused and duplicate samples, the final dataset comprises 4,289 images and 32,632 QA pairs (precise distribution in Table 1). For data preparation, images are resized to 224x224 pixels using BICUBIC interpolation to preserve critical pathological details, followed by normalization. Due to hardware constraints, a random subset of 3,000 samples was partitioned into an 80/20 training/validation split and loaded into PyTorch DataLoaders (batch size = 32), with shuffling enabled exclusively for training to prevent overfitting.

Table 1. Statistical distribution of the PathVQA dataset splits.

	Training Set	**Validation Set**	**Test Set**
QAs	19,654	6,259	6,719
Images	2,599	832	858

Figure 1. PathVQA Dataset Examples Exhibiting Selected QA Length Milestones. Illustrations depict individual cases for (i) Minimum QA length (15 tokens/characters; Index: 2093), (ii) Dataset average QA length (50 tokens/characters; Index: 282), and (iii) Maximum QA length (254 tokens/characters; Index: 6132), showing the paired medical imagery and QA text annotations.

Methodology

Four fusion approaches were evaluated within a GPT-2 decoder framework: Cross-Attention (CA), Gated Cross-Attention (Gated CA), Parallel Cross-Attention, and CLIP. The **Gated Cross-Attention** design extends standard CA by introducing a learned sigmoid gate applied element-wise to the cross-attention output before the residual addition. This gate modulates how much visual information is injected at each transformer block, selectively suppressing irrelevant image features, a mechanism especially relevant in medical imaging where questions may concern only localised pathological regions.

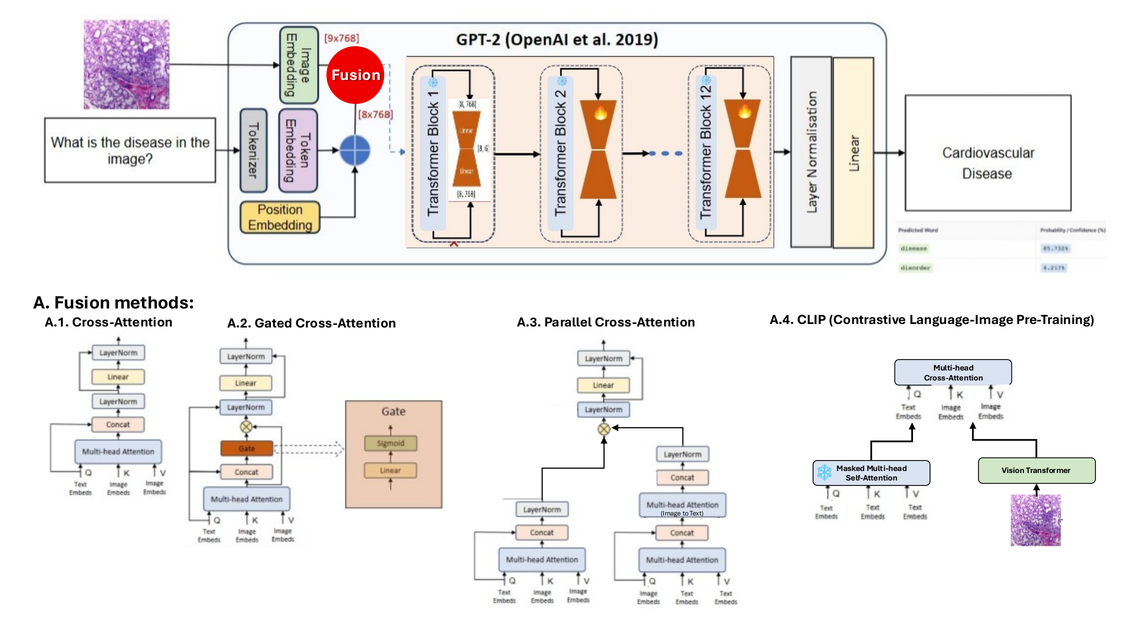


Figure 2. Model architecture for different fusion approaches.

The framework integrates an image encoder and text tokenizer with a GPT-2 language model decoder. Sub-panels illustrate four distinct feature fusion layers investigated during hyperparameter tuning: (A.1) Cross-Attention, (A.2) Gated Cross-Attention, (A.3) Parallel Cross-Attention, and (A.4) the CLIP (Contrastive Language–Image Pre-training) encoding configuration.

Results

Group A.

CA achieves the strongest qualitative accuracy with fully correct predictions (BLEU-1 = 0.3835, Rouge-L = 0.4730). Gated CA produces semantically adjacent but incorrect outputs (e.g., "heart" instead of "cardiovascular"), as the additional gate parameters require more data or training steps to converge, yielding slightly lower scores (BLEU-1 = 0.2114). Parallel CA overfits severely (BLEU-1 = 0.1112), while CLIP achieves competitive metrics but exhibits factual errors, indicating its contrastive pretraining objective misaligns with open-ended generative medical QA.

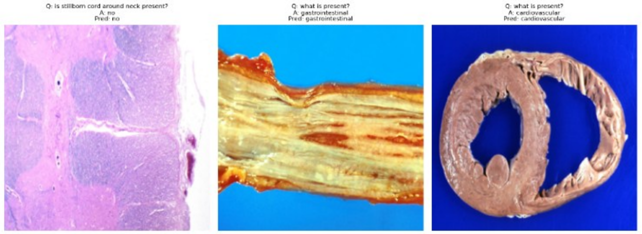
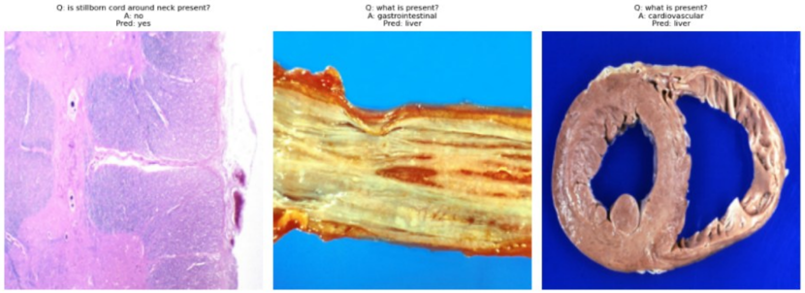
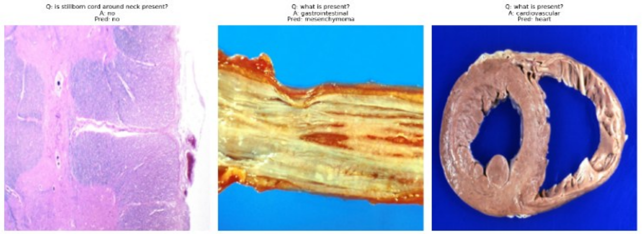
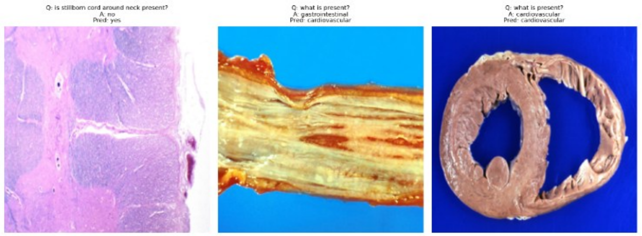
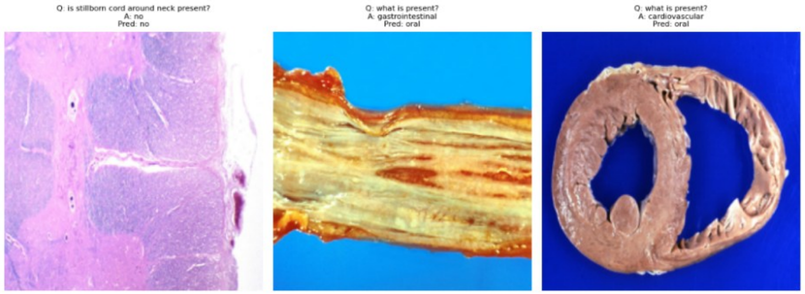
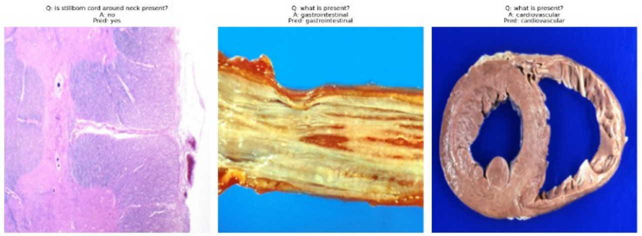
Group B.

Using Gated CA as baseline, three hyperparameter axes were explored. For **learning rate** (B-1), the default lr = 2×10^{-4} represents the optimal balance; a higher rate (lr = 2×10^{-3}) produces deceptively low loss (1.7532) but entirely incorrect predictions due to unstable weight updates that prevent meaningful cross-modal alignment, while an excessively low rate (lr = 2×10^{-5}) causes underfitting (loss = 3.4169) as the model fails to sufficiently update the LoRA adapters within the available epochs. For **LoRA rank** (B-2), reducing rank to 4 improves prediction correctness over the default rank 8, because a smaller parameter subspace forces the model to learn only the most task-relevant low-rank adaptations and discards noise-inducing redundancies; rank 16 introduces sufficient over-parameterisation to incur one prediction error despite higher capacity. For **attention heads** (B-3), scaling to 8 heads yields the highest Rouge-1 (0.5145) and Meteor (0.2610), as finer-grained parallel attention sub-spaces improve syntactic fluency; however, both 2 and 8 heads still incur one qualitative error, demonstrating that head count primarily governs output fluency rather than cross-modal factual grounding.

Conclusion

Quantitative metrics and qualitative samples must be interpreted jointly in medical VQA. The optimised Gated CA (8 attention heads) surpasses CA and CLIP on Rouge-1 and Meteor, demonstrating that the gating mechanism realises its advantage only after targeted tuning, particularly smaller LoRA rank and appropriate learning rate, which together stabilise cross-modal feature integration and constrain unnecessary parameter growth.

Student ID: 14210037

<table>
<tr><th colspan="2">(a) Comparison of different multi-modal fusion methods (Group A)</th><th colspan="2">(b) Impact of different learning rates (Group B1)</th></tr>
<tr><td>(A.1)
Cross-Attention</td><td>
</td><td rowspan="2">Lr= 2x10^-3</td><td rowspan="2">
</td></tr>
<tr><td>(A.2) Gated
Cross-Attention</td><td>
</td></tr>
<tr><td>(A.3) Parallel
Cross-Attention</td><td>
</td><td rowspan="2">Lr= 2x10^-5</td><td rowspan="2">
</td></tr>
<tr><td>(A.4) CLIP</td><td>
</td></tr>
</table>

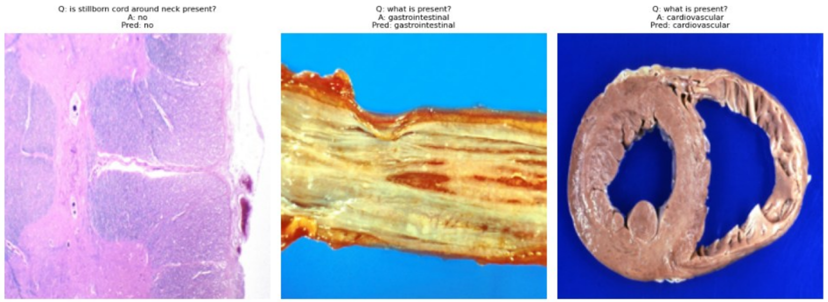
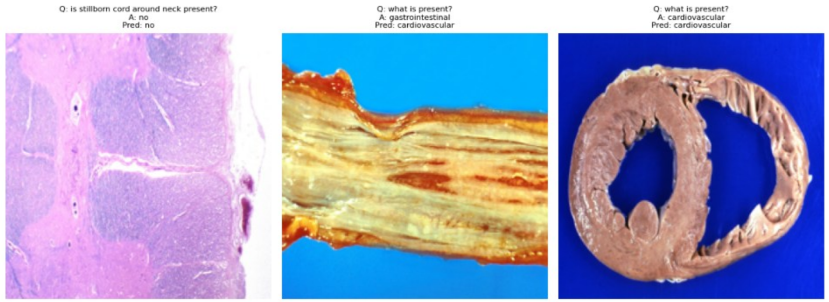
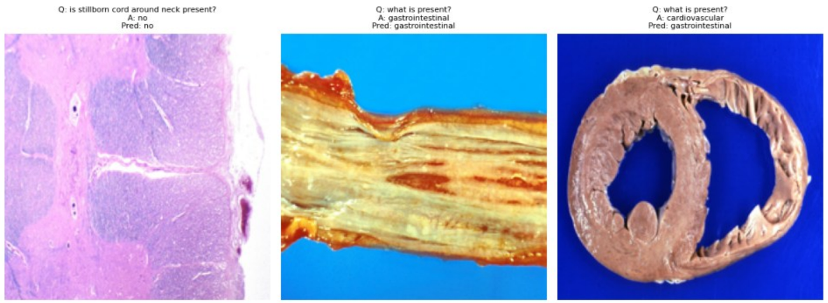
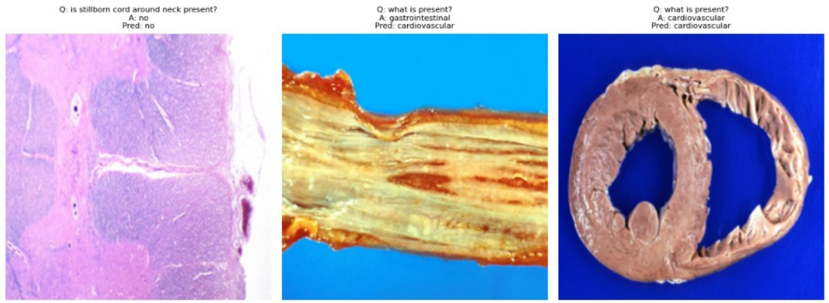
<table>
<tr><th colspan="2">(c) Effects of varying LoRA ranks (Group B2)</th><th colspan="2">(d) Influence of the number of cross-attention heads (Group B3)</th></tr>
<tr><td>LoRA = 4</td><td>
</td><td>number of attention heads = 2</td><td>
</td></tr>
<tr><td>LoRA = 16</td><td>
</td><td>number of attention heads = 8</td><td>
</td></tr>
</table>

Figure 3. Qualitative comparison of multi-modal fusion methods and hyperparameter configurations.

(a) Comparison of different multi-modal fusion methods (Group A): Visual results of three selected samples across four configurations: (A.1) Cross-Attention yields fully correct predictions, demonstrating high accuracy; (A.2) Gated Cross-Attention produces an incorrect yet semantically relevant prediction (e.g., predicting "heart" instead of "cardiovascular"); (A.3) Parallel Cross-Attention and (A.4) CLIP exhibit sub-optimal performance with notable errors. **(b) Impact of different learning rates (Group B1):** Visual comparison of model predictions under varying learning rates, where a lower rate delivers superior prediction accuracy compared to a higher rate. **(c) Effects of varying LoRA ranks (Group B2):** Visual evaluation of model performance across different LoRA ranks, demonstrating that a smaller rank size contributes to improved prediction outcomes. **(d) Influence of the number of cross-attention heads (Group B3):** Visual representation of predictions across different head counts, illustrating that variations in attention heads yield no significant difference in performance.

Table 2. Quantitative text generation evaluation and loss performance across varying multimodal fusion mechanisms and hyperparameter configurations. BLEU, Rouge-1, Rouge-L, and Meteor are standard evaluation metrics for text generation quality, where higher values indicate better performance. Model 2 (Gated Cross-Attention) serves as the control baseline for all hyperparameter tuning sub-experiments in Group B.

		Best Average Training Loss	BLEU-1	BLEU-2	Rouge1	RougeL	Meteor
Group A: Architectural Fusion Variants							
1	Cross-attention	2.2202	0.3835	0.0284	0.4749	0.4730	0.2393
2	Gated Cross-Attention	1.7645	0.2114	0.0040	0.4361	0.4345	0.2207
3	Parallel Cross-Attention	1.6369	0.1112	0.0029	0.3229	0.3238	0.1664
4	CLIP	2.1770	0.3793	0.0273	0.4949	0.4931	0.2516
Group B: Hyperparameter Tuning (Baseline: Gated Cross-Attention)							
B-1. Learning rate: default = 2x10^-4 (lr=0.0002).							
5	2x10^-3 (lr=0.002)	1.7532	0.3221	0.0127	0.4641	0.4656	0.2341
6	2x10^-5 (lr=0.00002)	3.4169	0.2878	0.0000	0.2939	0.2933	0.1469
B-2. LoRA: default = 8							
7	LoRA = 4	2.4327	0.3151	0.0134	0.4472	0.4467	0.2253
8	LoRA = 16	2.0155	0.3133	0.0158	0.4388	0.4366	0.2218
B-3. cross-attention heads: default = 4							
9	num_head fussion = 2	1.9781	0.3989	0.0207	0.4699	0.4679	0.2360
10	num_head fussion = 8	2.2147	0.3538	0.0203	0.5145	0.5150	0.2610

Reference

Amit, Hey. “Implement Self-Attention and Cross-Attention in Pytorch.” *Medium*, 7 Nov. 2024, medium.com/@heyamit10/implement-self-attention-and-cross-attention-in-pytorch-cfe17ab0b3ee.

DeepWiki. “GatedCrossAttentionBlock.” *DeepWiki*, 23 Apr. 2025, deepwiki.com/lucidrains/flamingo-pytorch/3.2-gatedcrossattentionblock. Accessed 25 May 2026.

Hu, Jian, et al. “SpaGCN: Integrating Gene Expression, Spatial Location and Histology to Identify Spatial Domains and Spatially Variable Genes by Graph Convolutional Network.” *Nature Methods*, vol. 18, no. 11, 28 Oct. 2021, pp. 1342–1351, https://doi.org/10.1038/s41592-021-01255-8.

OpenAI. “CLIP: Connecting Text and Images.” *Openai.com*, 2021, openai.com/index/clip/.

openai. “CLIP/Clip/Model.py at Main · Openai/CLIP.” *GitHub*, 2025, github.com/openai/CLIP/blob/main/clip/model.py. Accessed 25 May 2026.

Wikipedia Contributors. “Contrastive Language-Image Pre-Training.” *Wikipedia*, Wikimedia Foundation, 5 Dec. 2024, en.wikipedia.org/wiki/Contrastive_Language-Image_Pre-training.